Q1) (15 pts.) What is the reverse complement:

a) AGCTACGTCG TCGATGCAGC → CGACGTAGCT

b) CTTGCAATGCT GAACGTTACGA → AGCATTGCAAG

c) ATGCGCCA TACGCGGT → TGGCGCAT

Q2) (15 pts.) The d-neighborhood of a k-mer Pattern is the collection of all k-mers that are at most Hamming distance d from Pattern. How many 4-mers are in 3-neighborhood of Pattern =/ATGC? Note that the d-neighborhood of pattern includes Pattern.

Q3) (15 pts.) Find the number of occurrences of pattern **AGCT** with 1 mismatch / 2 mismatches in **CGCTAGATATAGCTCAGCTA**

CGCT
GCTA

Q4) (25 pts.) Assume that we will apply Randomized Motif Search Algorithm to the following DNA strings

TGACGTTC

TAAGAGTT

GGACGAAA

CTGTTCGC

Assume that the following 3-mers are selected randomly:

TGA

GTT

GAA

TGT

a) (8 pts.) What is score(motifs)? What is the minimum and maximum possible value of *Score(Motifs)* for 4 motifs of length 3?

Current score = 2 + 2 + 2 = 6 ✓

Min score = 0

Max score = 9

b) (4 pts.) What is profile(motifs)?

A	0	0.25	0.50
T	0.5	0.25	0.50
G	0.5	0.50	0
C	0	0	0

✓

c) (8 pts.) What are the new motifs after one iteration is completed?

TGA
TAA
GGA
TGT

Score: 1 + 1 + 1 = 3 ✓

d) (5 pts.) Do the new motifs represent the consensus string better than the initial motifs? Compare the previous and new motifs.

Consensus string(s) = TGACG{AT/GC}T{C/T}

Yes, because we choose high probability motifs after each iteration

Q5) (30 pts.) If we construct the de Bruijn Graphs for the following:

A) DeBruijn3(ATCGTGTAATGTCA)

B) DeBruijn3(ATCGTGTCATGTAA)

Are they the same or different, why? (DeBruijnk: k denotes the length of k-mers used to construct the graph. Here 3-mers are going to be used)

A) ATC, TCG, CGT, GTG, TGT, GTA, TAA, AAT, ATG, TGT, GTC, TCA

B) ATC, TCG, CGT, GTG, TGT, GTC, TCA, CAT, ATG, TGT, GTA, TAA

Due to they do not have the same 3-mers those graphs are not the same.